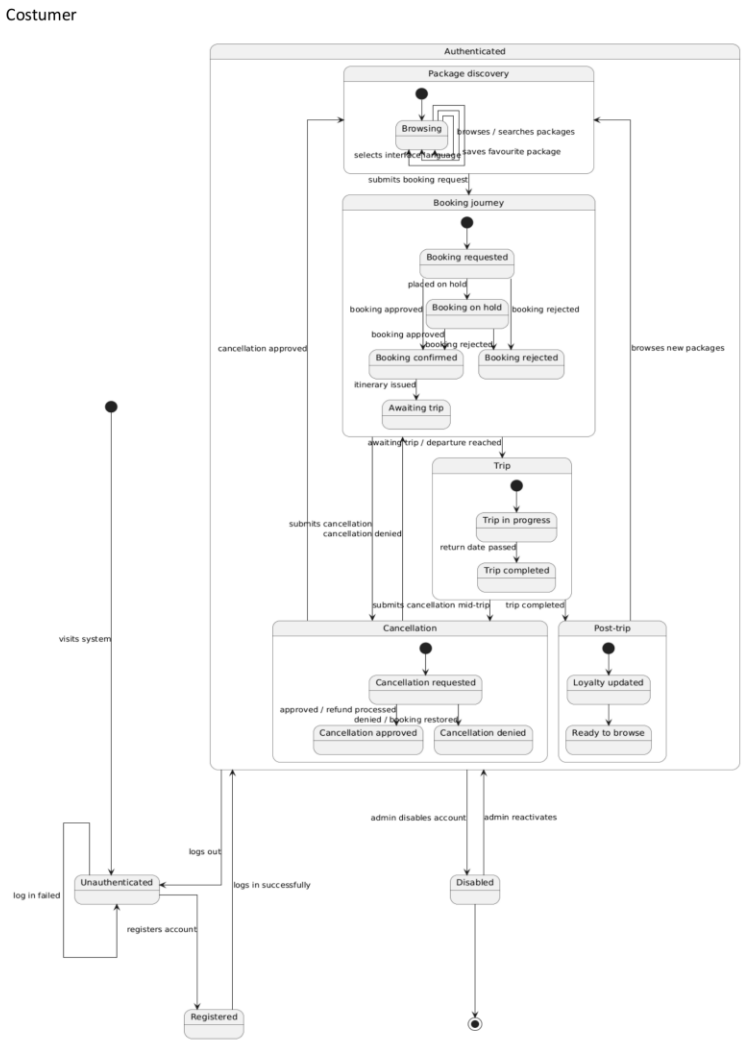
State Machine Diagrams

1. Customer

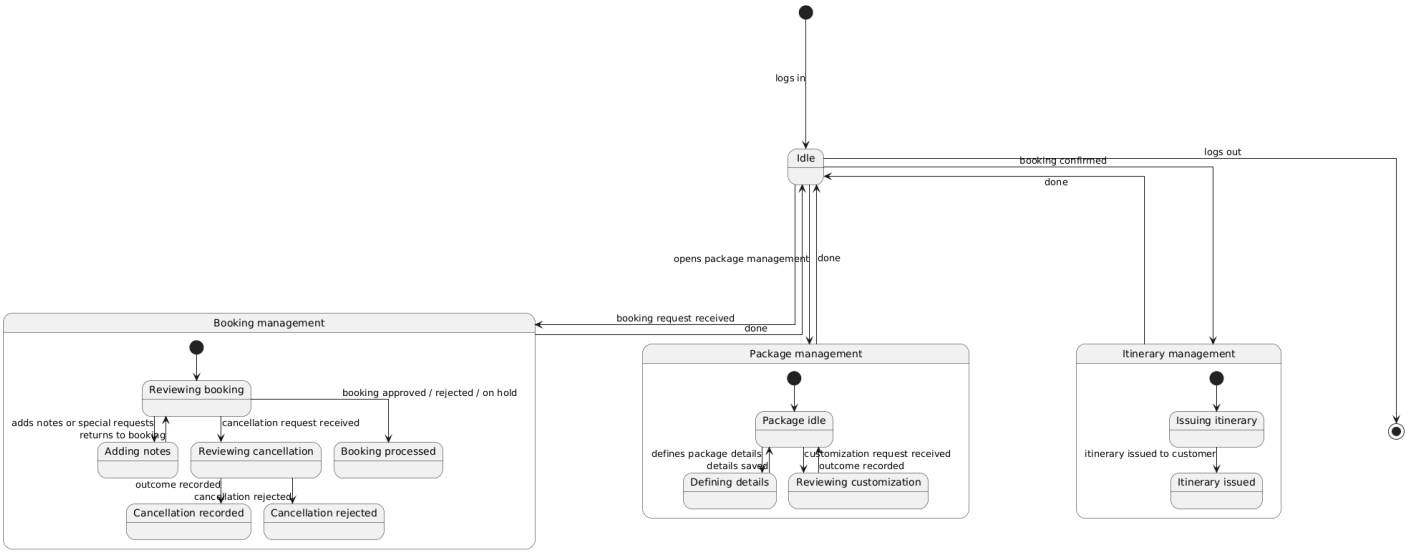
The Authenticated composite state is subdivided into five clearly separated regions: Package Discovery, Booking Journey, Trip, Cancellation, and Post-Trip. This separation reflects the distinct phases a customer moves through and prevents the diagram from becoming a tangled web of cross-state transitions.



2. Travel Agent

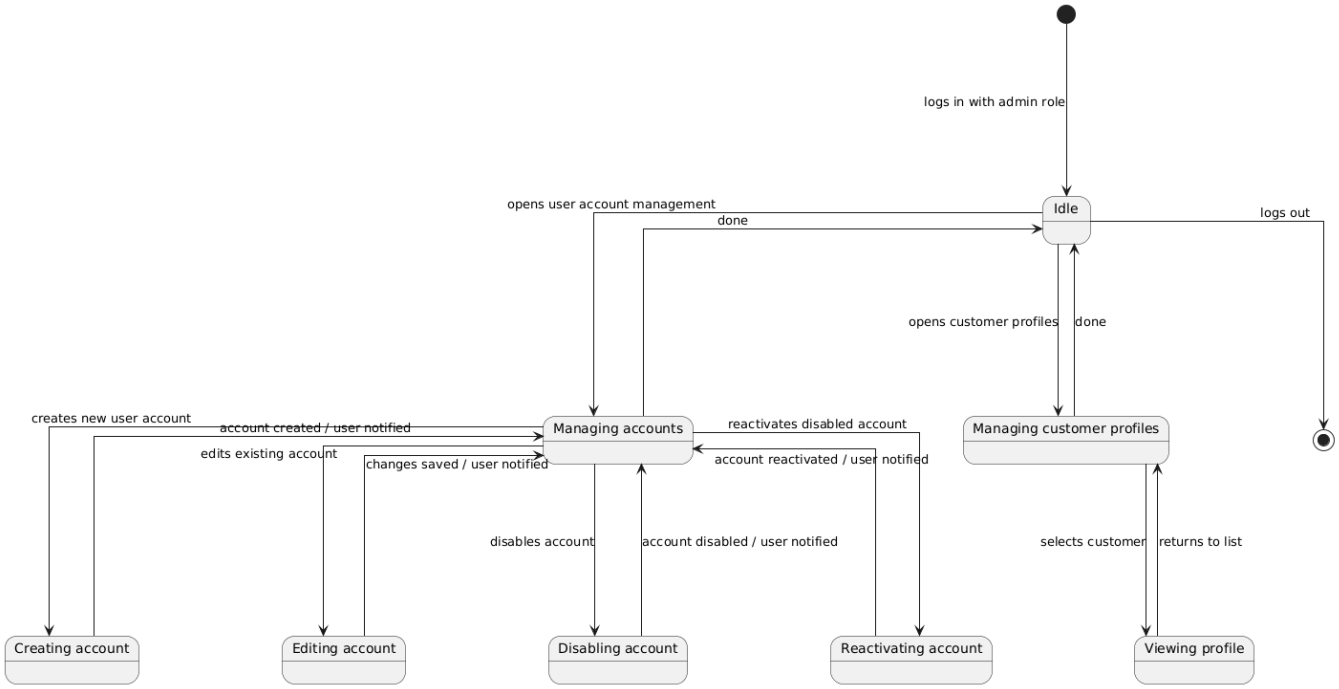
The Travel Agent is responsible for processing booking requests, handling cancellation reviews, managing packages, and issuing digital itineraries to confirmed customers. Rather than routing every transition through a single Idle state, this diagram groups related work into three composite regions: Booking Management, Package Management, and Itinerary Management.

The Idle state acts as a clean hub connecting only to the three work regions and the session exit, keeping arrow labels readable and uncluttered.



3. Administrator

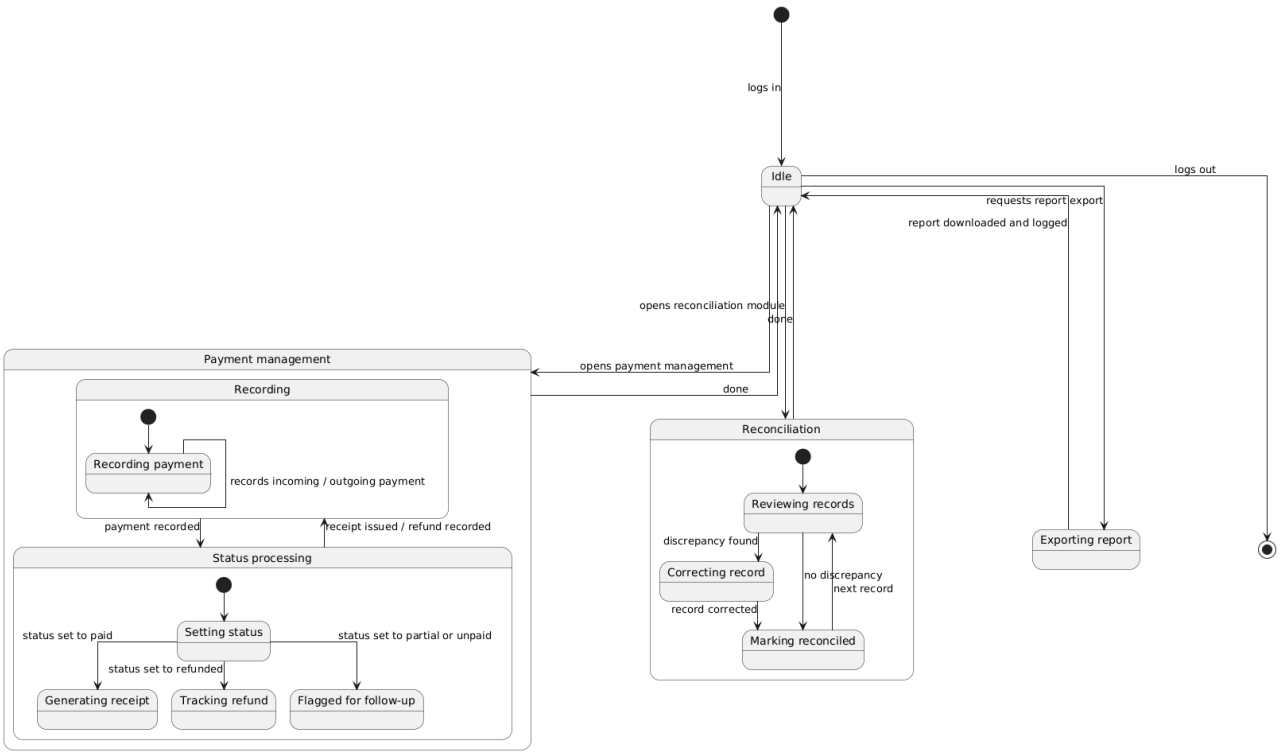
The Administrator manages user accounts and customer profiles. This diagram covers the four account lifecycle actions — create, edit, disable, and reactivate — as well as customer profile access. All actions return to the Idle state upon completion, with user notification modeled as part of each transition.



4. Staff

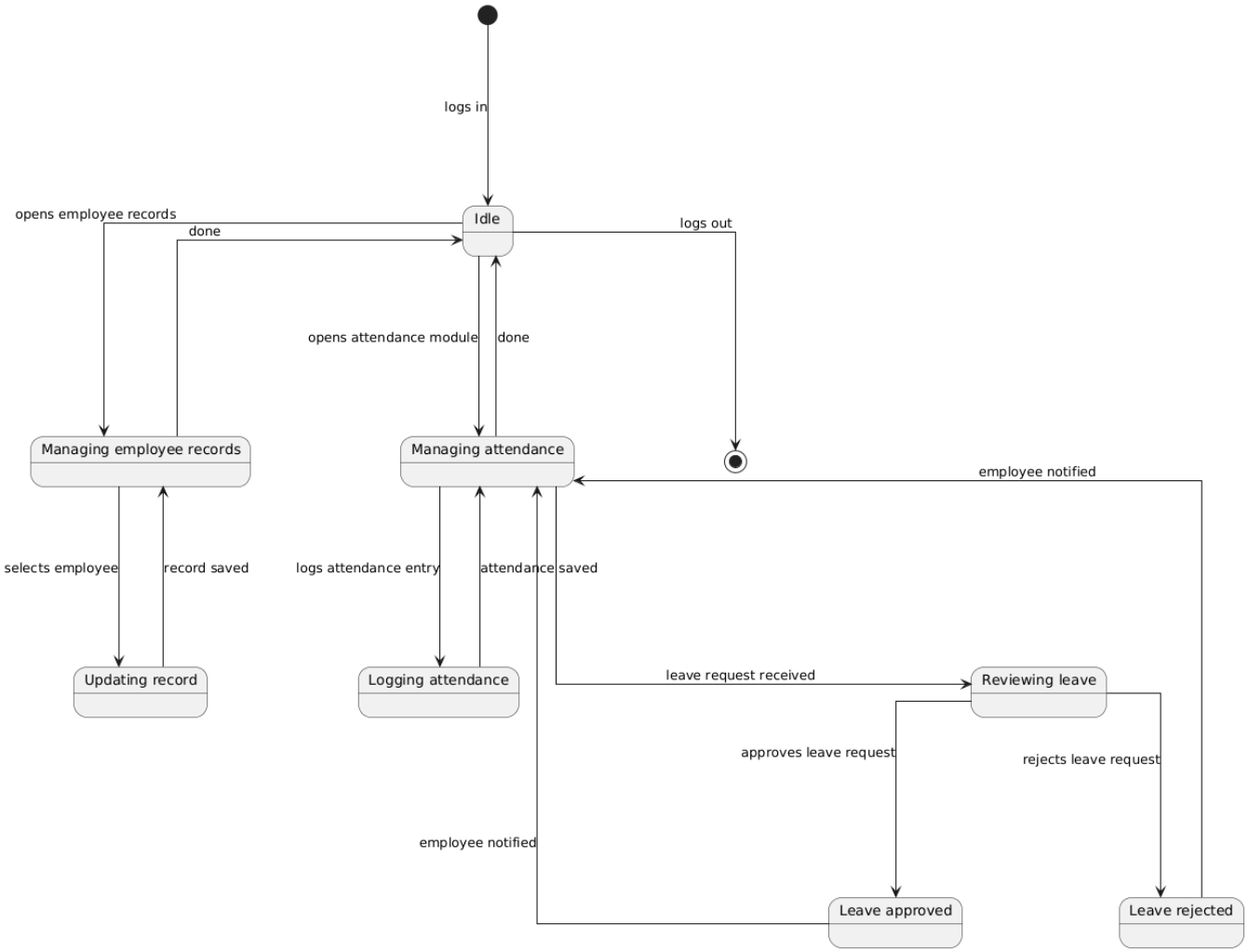
Staff is the broadest operational actor in the system, responsible for travel package management, scheduling and trip operations, partner management, and system audit logging. The diagram uses nested composite states to separate concerns: Managing Packages is subdivided into Package Catalogue and Package Configuration, while Managing Schedules is subdivided into Schedule Maintenance and Trip Closure

This two-level nesting keeps related actions together and prevents the arrow density that would result from a flat design.



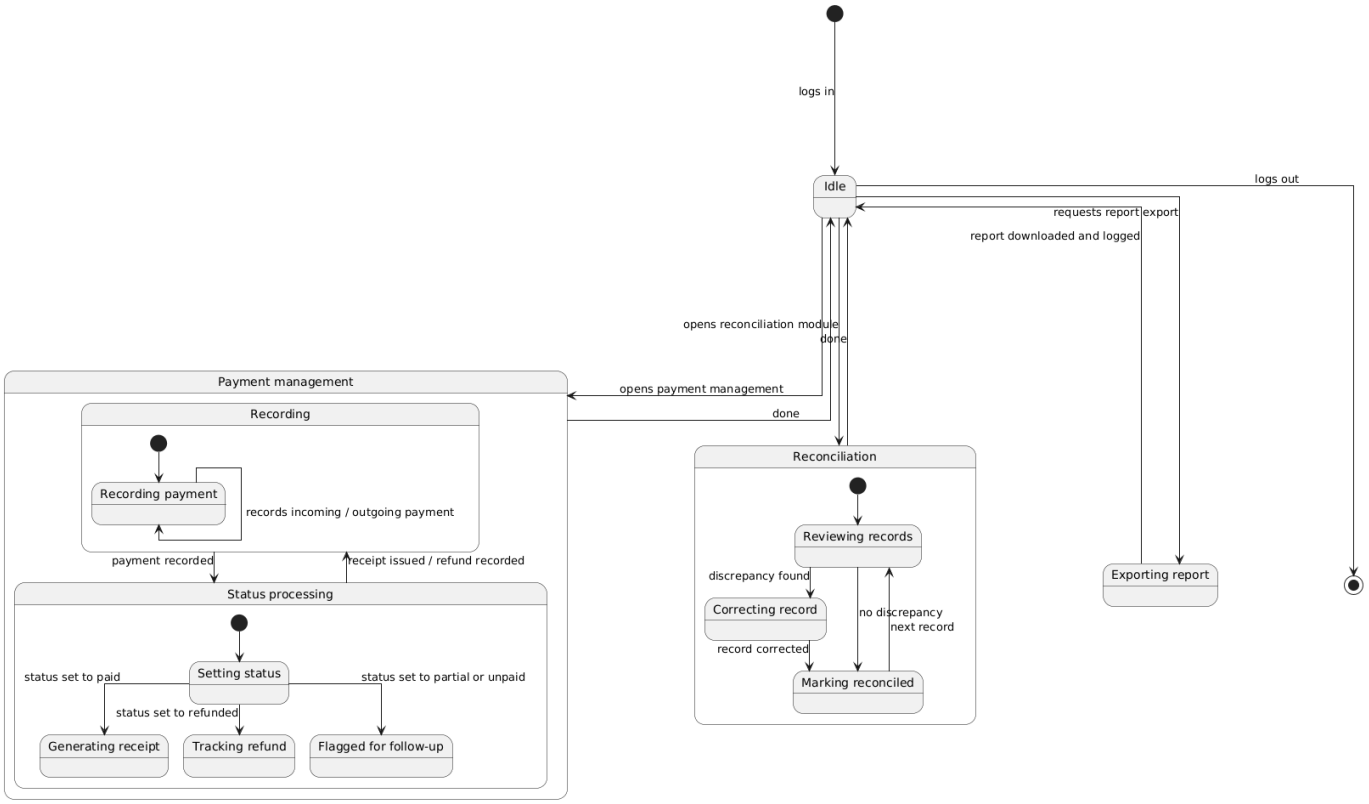
5. HR Staff

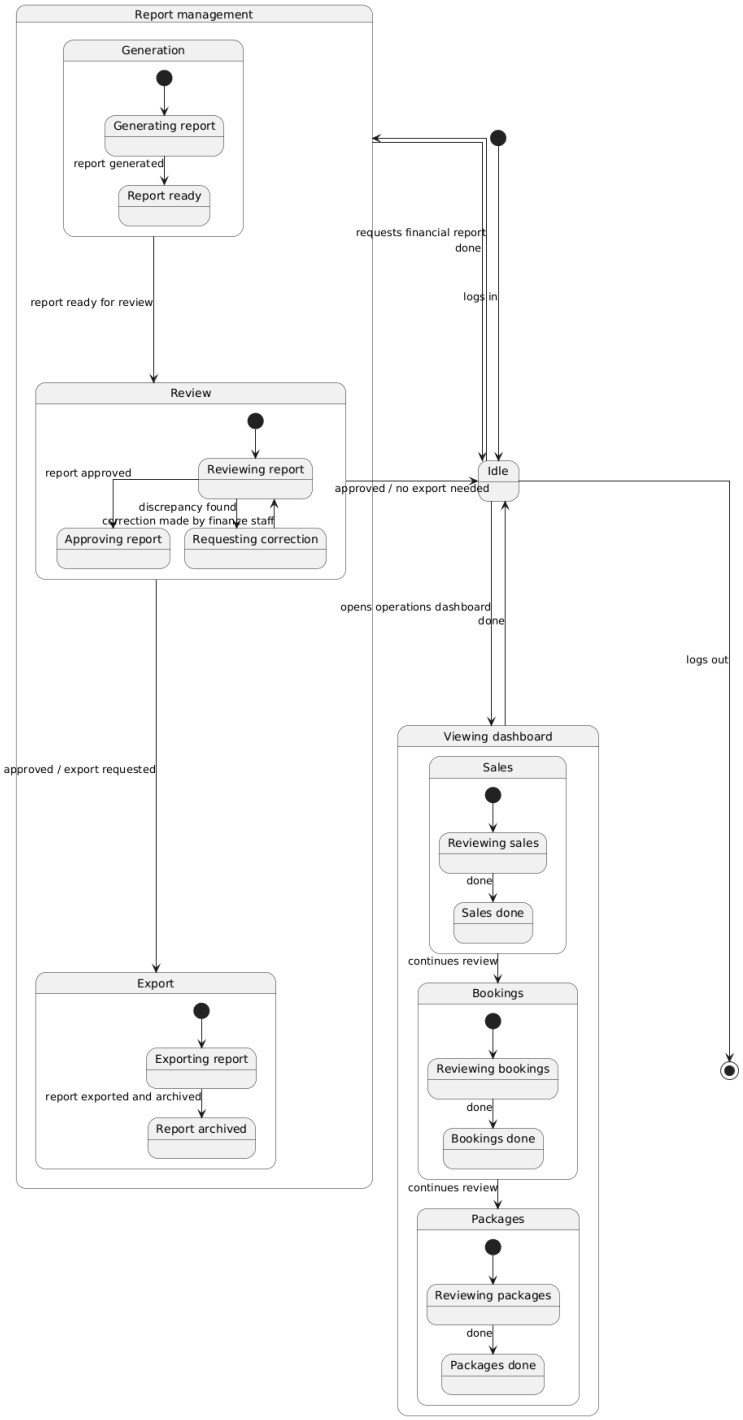
HR Staff manage employee records and the attendance and leave approval workflow. The diagram covers two parallel work areas accessible from Idle: Managing Employee Records, which handles individual record updates, and Managing Attendance, which covers attendance logging and leave request review with approve and reject outcomes.



6. Finance Staff

Finance Staff handle all payment operations and financial record management. This diagram groups work into three composite regions: Payment Management (subdivided into Recording and Status Processing), Reconciliation, and Report Export. The separation of Recording from Status Processing eliminates the cluttered back-and-forth transitions that would otherwise appear between those two states.



7. Manager / Owner

The Manager oversees agency performance through the operations dashboard and financial report management. The Viewing Dashboard composite state is subdivided into three sequential review regions Sales, Bookings, and Packages so that each review area has its own clear flow. Report Management is subdivided into Generation, Review, and Export, with a correction loop back to Review when a discrepancy is found.